



SGL2: Pathology of the Bowel

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MBChB / FIBMS-Pathology

Ischemic Bowel Disease

- Infarction of bowel due to hypoperfusion.
- It is mainly due to impair arterial supply but uncommonly is due to impair venous drainage.
- **Causes of decrease arterial supply:**
 - Non obstructive hypoperfusion e.g. hypovolemic shock.

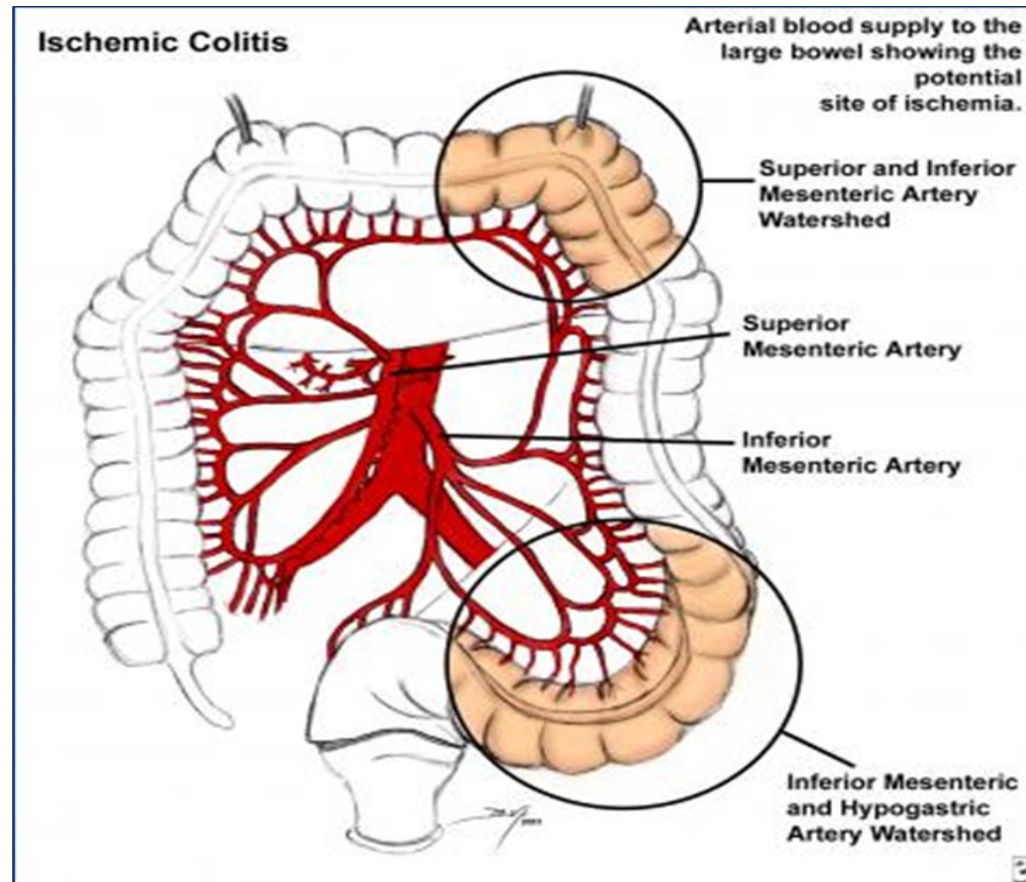
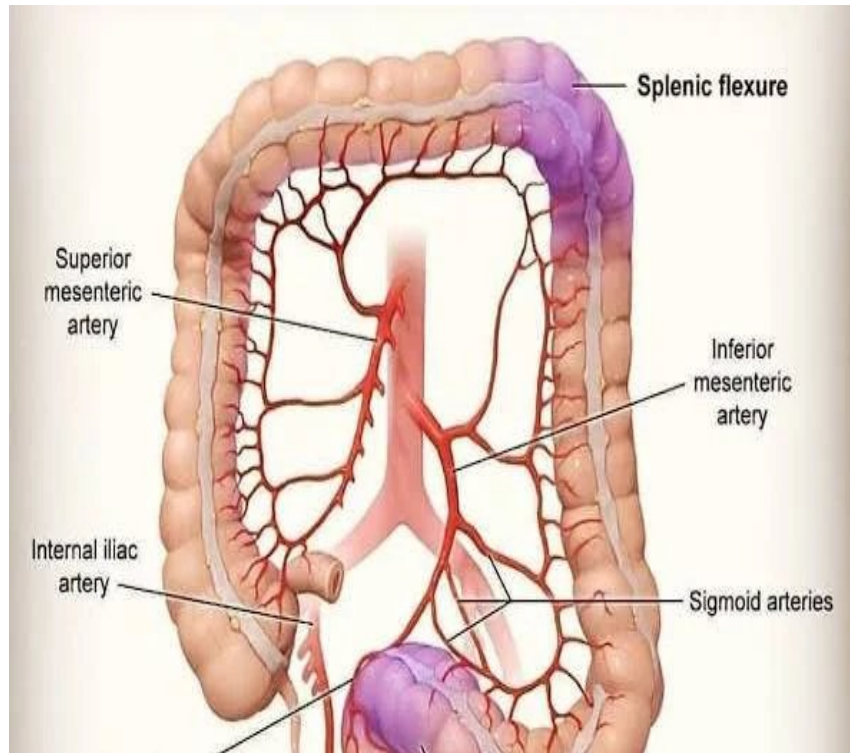
- Arterial obstruction causes include:
 - Severe atherosclerosis.
 - Aortic aneurysm
 - Hypercoagulable states
 - Oral contraceptive use.
 - Embolization of cardiac vegetations or aortic atheroma.
 - Vasoconstrictive drugs
 - Systemic vasculitides.

■ Causes of impaired venous drainage:

- Mesenteric venous thrombosis.
- Other causes include: invasive neoplasms, portal hypertension, trauma.

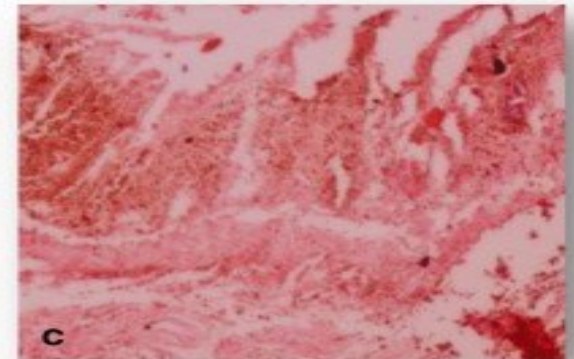
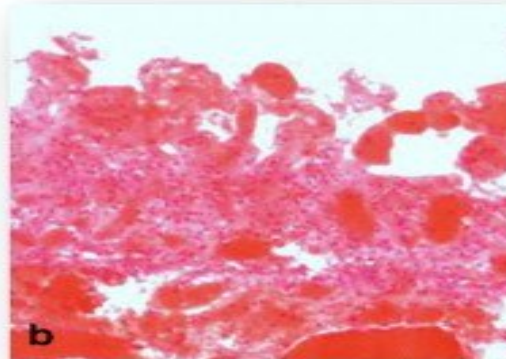
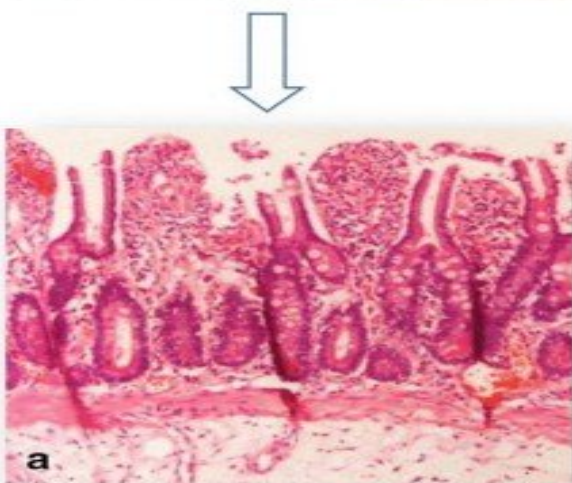
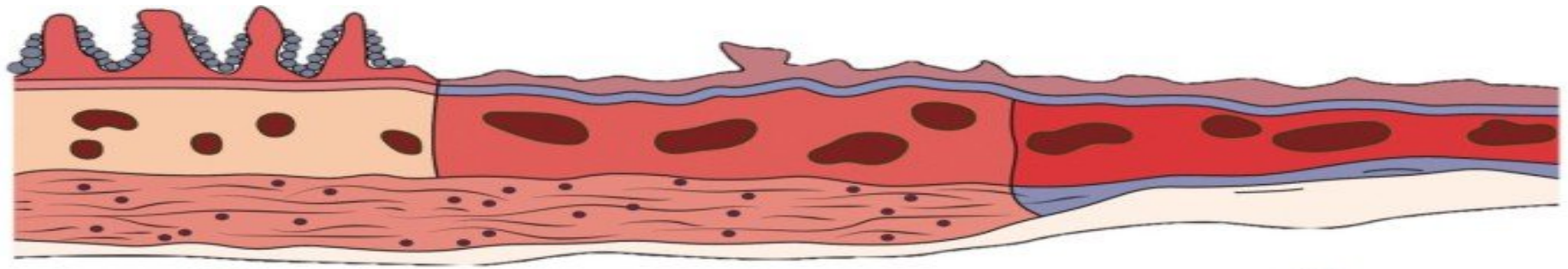
■ Watershed zones are most liable to ischemia:-

- Splenic flexure
- Sigmoid colon



■ Morphology:

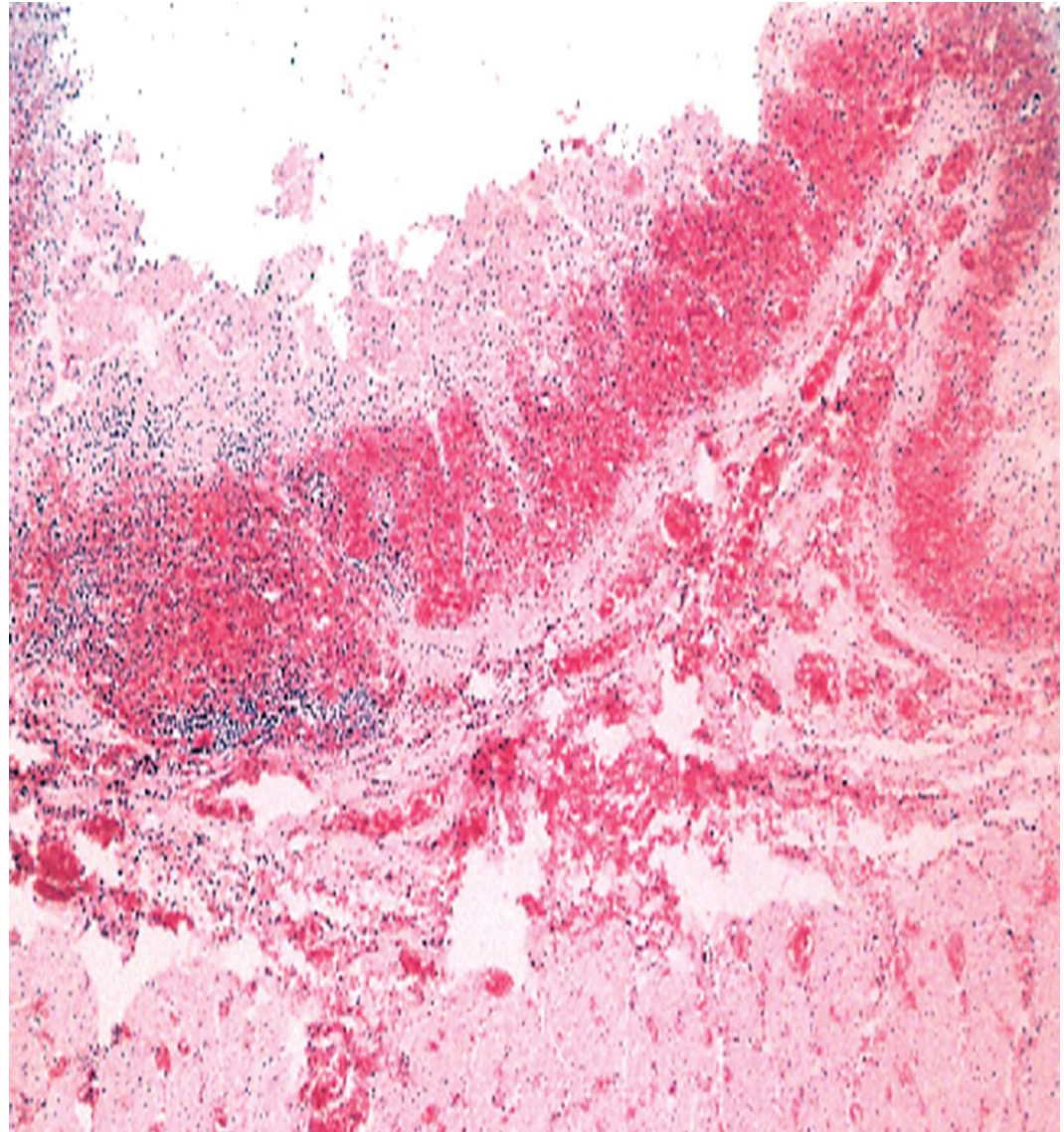
- Mucosal infarction (a)
- Mural infarction of mucosa and submucosa (b).
- Transmural infarction involving all three layers (c).



Gross Pathology



Microscopic pathology



Malabsorption syndrome

Is the inability to absorb nutrients, vitamins, and minerals from the intestinal tract into bloodstream.

Clinical presentation:

- General features: Diarrhea, Steatorrhea, flatus, abdominal pain, and weight loss.
- Symptoms associated with specific deficiencies:-
 - Anemia: iron, folate, Vitamins B6, B12
 - Bleeding / purpura: Vitamin K
 - Osteopenia, tetany: calcium, magnesium, vitamin D
 - Edema: albumin.....etc.

Physiology of Absorption:

- Disturbances are related to:
 - Intraluminal digestion: saliva, gastric, pancreatic, small bowel & bile salts secretion.
 - Mucosal (terminal) digestion: hydrolysis of complex carbohydrates, protein, and fat in brush border of small bowel.
 - Transepithelial and Lymphatic transport: across small bowel epithelium to intestinal vasculature.

Defects in Malabsorptive and Diarrheal Disease

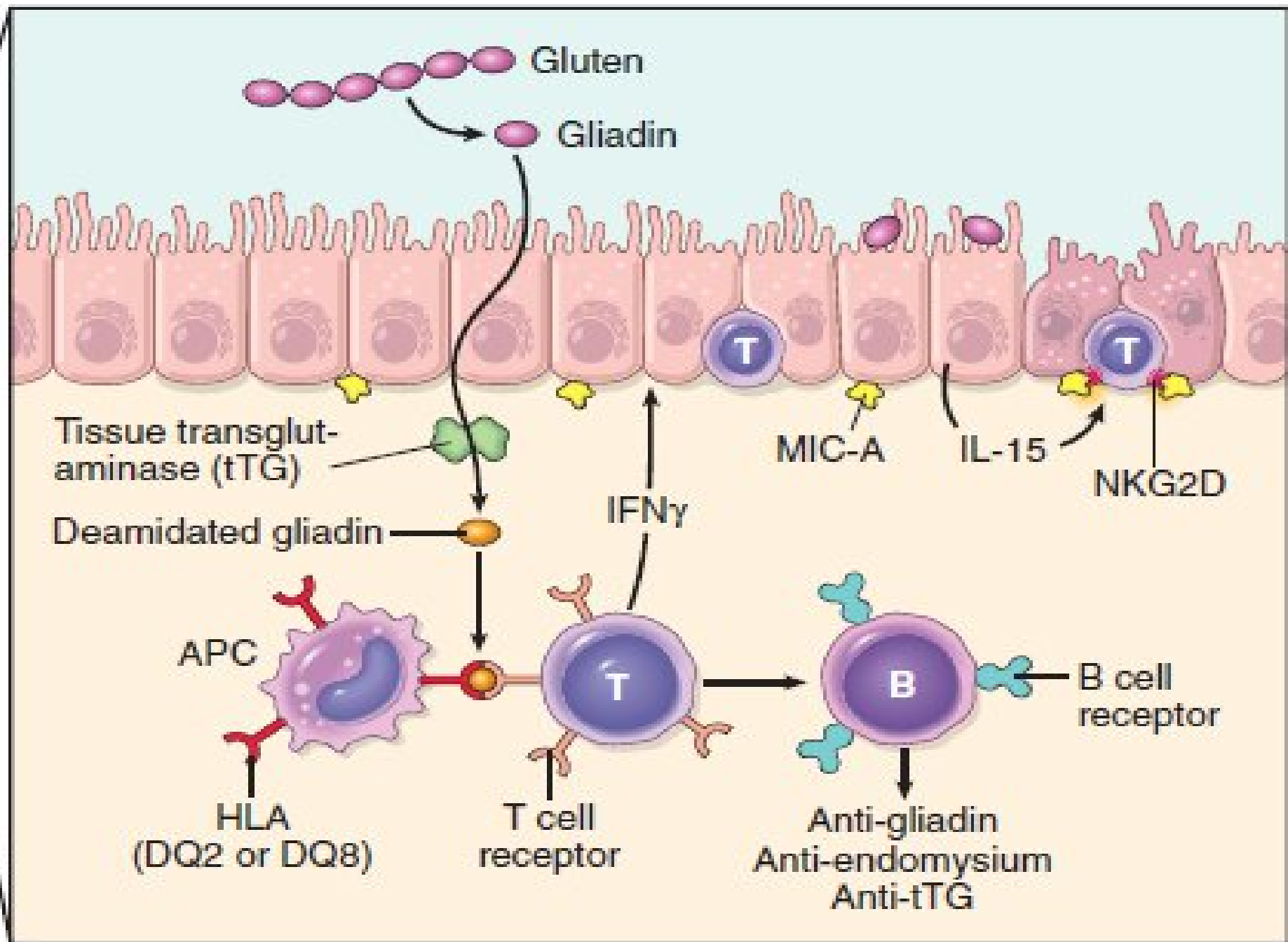
Disease	Intraluminal Digestion	Terminal Digestion	Transepithelial Transport	Lymphatic Transport
Celiac disease		+	+	
Tropical sprue		+	+	
Chronic pancreatitis	+			
Cystic fibrosis	+			
Primary bile acid malabsorption	+		+	
Autoimmune enteropathy		+	+	
Disaccharidase deficiency		+		
Whipple disease				+
Abetalipoproteinemia			+	
Viral gastroenteritis		+	+	
Bacterial gastroenteritis		+	+	
Parasitic gastroenteritis		+	+	
Inflammatory bowel disease	+	+	+	

Celiac Disease:

- Also known as non tropical sprue or gluten-sensitive enteropathy.
- Immune-mediated enteropathy triggered by ingestion of gluten-containing cereals such as wheat, rye, or barley in genetically predisposed persons.

Pathogenesis:

- Gluten → gliadin → deamidated by tissue transglutaminase → (in genetically susceptible person) → interact with HLA-DQ2 or HLA-DQ8 on antigen-presenting cells → presented to CD4+ T cells → CD 4+ T cells produce cytokines → tissue damage.
- The CD8+ cells are triggered and activated by enterocyte secreted IL15 → express MIC-A receptor NKG2D → so become cytotoxic and kill enterocytes.
- B cell response follows by producing anti-tissue transglutaminase, anti-deamidated gliadin & anti-endomysial antibodies which are diagnostically useful.



Morphology:

- Biopsy specimens from second portion of duodenum or proximal jejunum are generally diagnostic in celiac disease.
- Histology is characterized by:
 - Increased numbers of intraepithelial CD8+ T lymphocytes.
 - Crypt hyperplasia.
 - Villous atrophy.

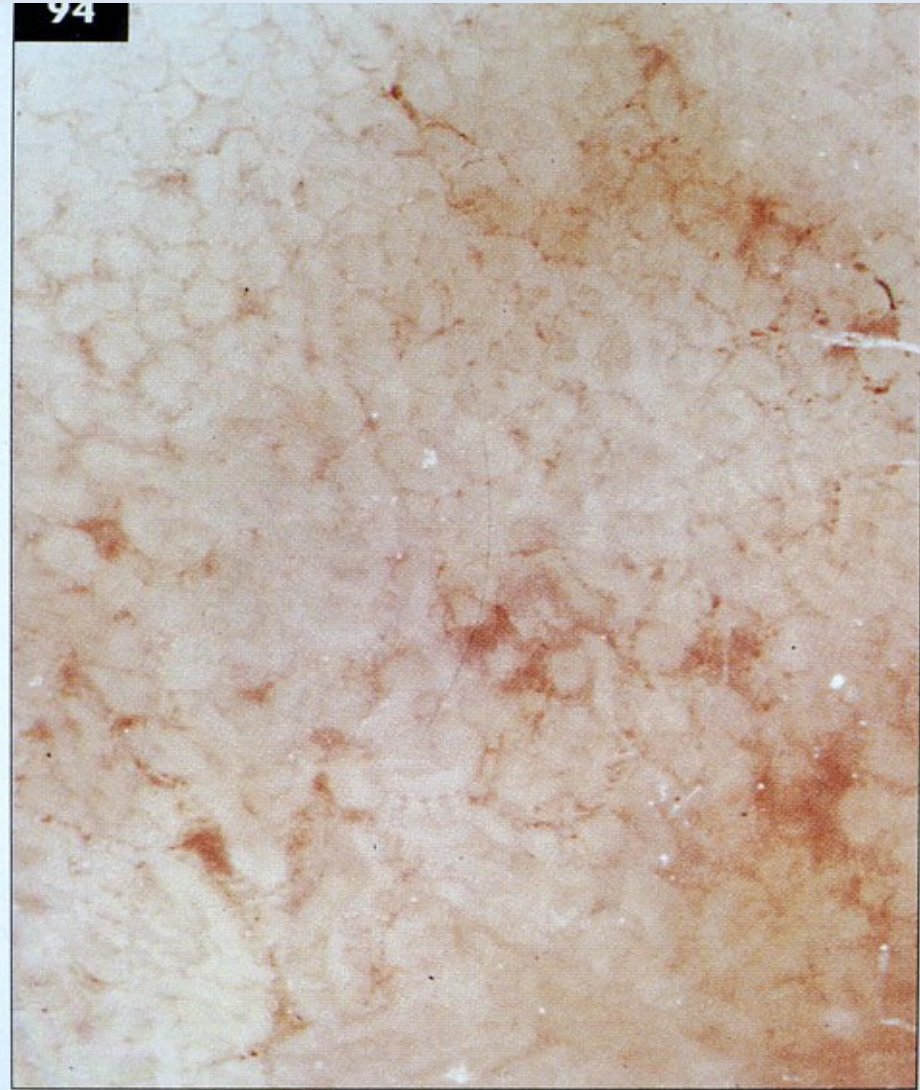
Comparing two small bowel biopsies:

87

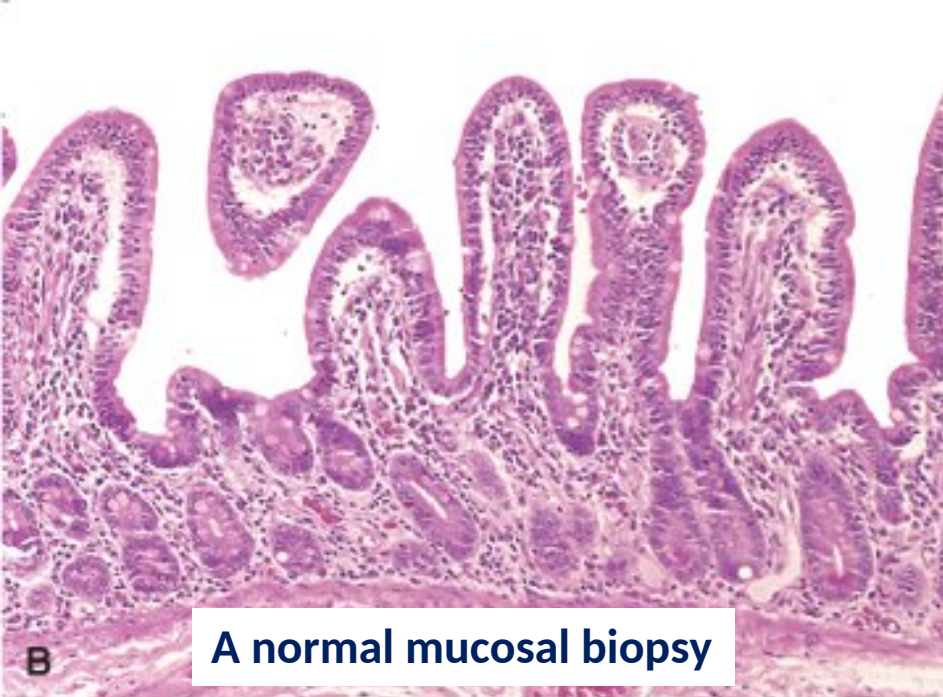


87 Normal dissection microscope appearances of finger-like villi.

94



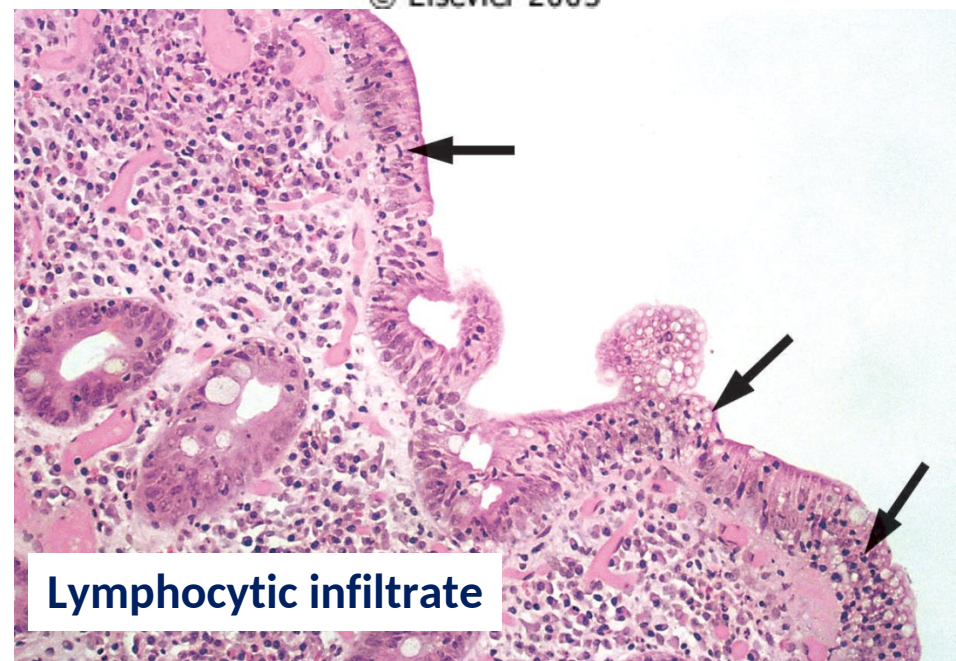
94 Flat jejunal biopsy under dissection microscope showing no villi and only crypts in coeliac disease.



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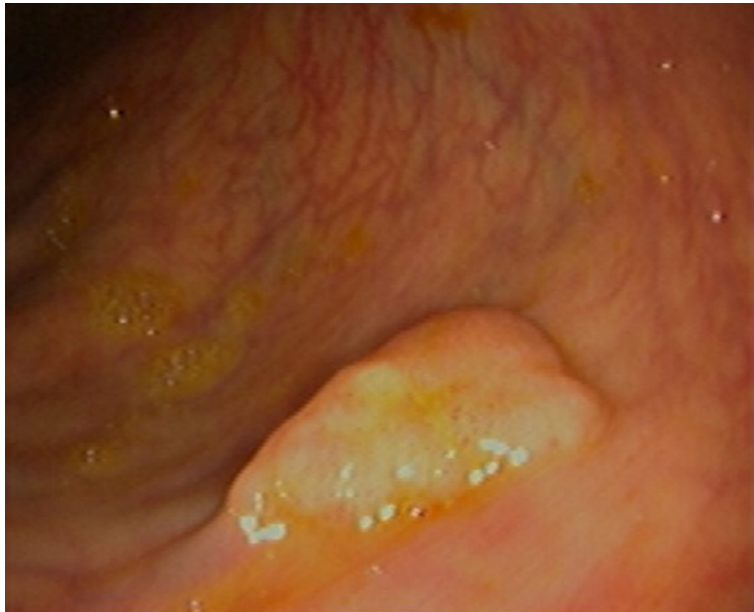
Clinical features:

- Infants: Failure to thrive, diarrhea.
- Adults: Diarrhea, flatulence, malabsorption, weight loss, and fatigue.
- Celiac disease is associated with:
 - Dermatitis herpetiformis.
 - Enteropathy-associated T cell lymphoma:- aggressive tumor of intraepithelial T lymphocytes.
 - Increase frequency of small intestinal adenocarc.
 - Splenic atrophy: for unknown reason.

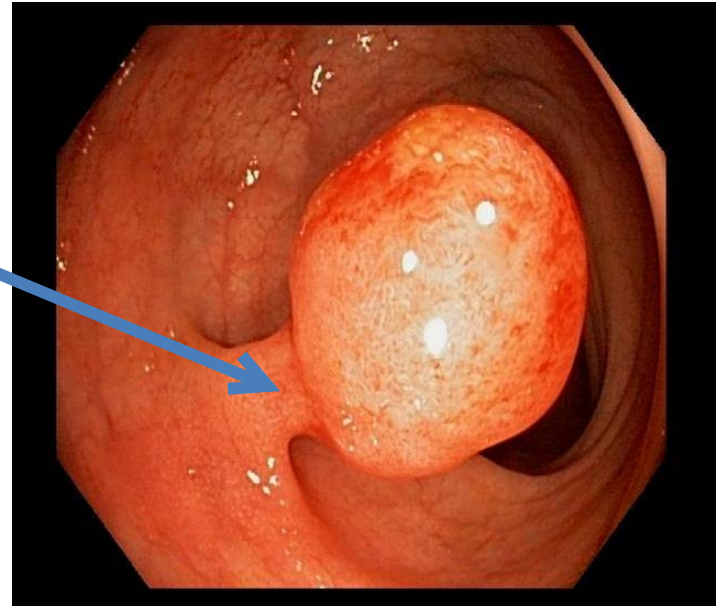
Treatment: is a gluten free diet.

Intestinal polyps

- Most common in the colon.
- Grossly:
 - Sessile polyps : those without stalks.
 - Pedunculated polyps: have stalks.



Stalks



- Intestinal polyps can be:
 - **Nonneoplastic:**
 - Inflammatory
 - Hamartomatous
 - Hyperplastic
 - **Neoplastic:** The most common is **adenomatous polyp**.

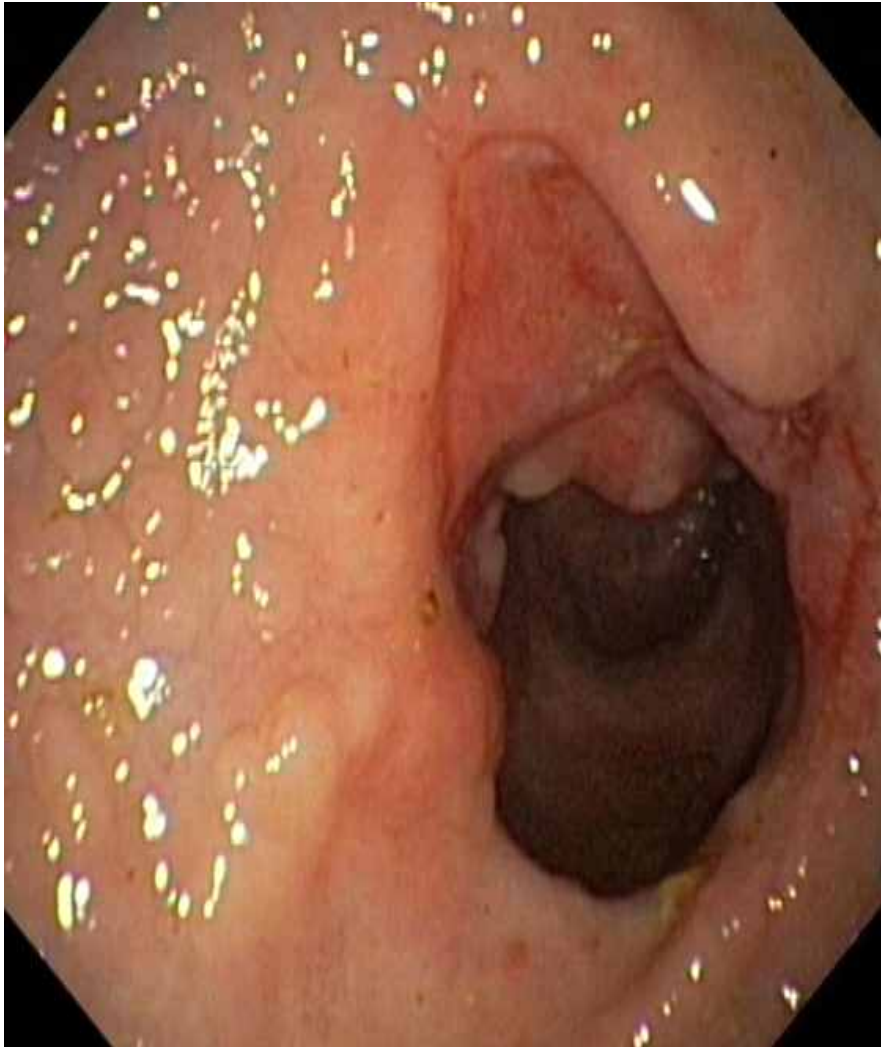
Nonneoplastic:

Inflammatory Polyps:

Seen in **Solitary Rectal Ulcer Syndrome:**

- Patients present with triad of rectal bleeding, mucus discharge, and an inflammatory lesion of anterior rectal wall.
- Is due to impaired relaxation of anorectal sphincter.
- Chronic cycles of injury and healing produce a polypoid mass.

Solitary Rectal Ulcer with polyp

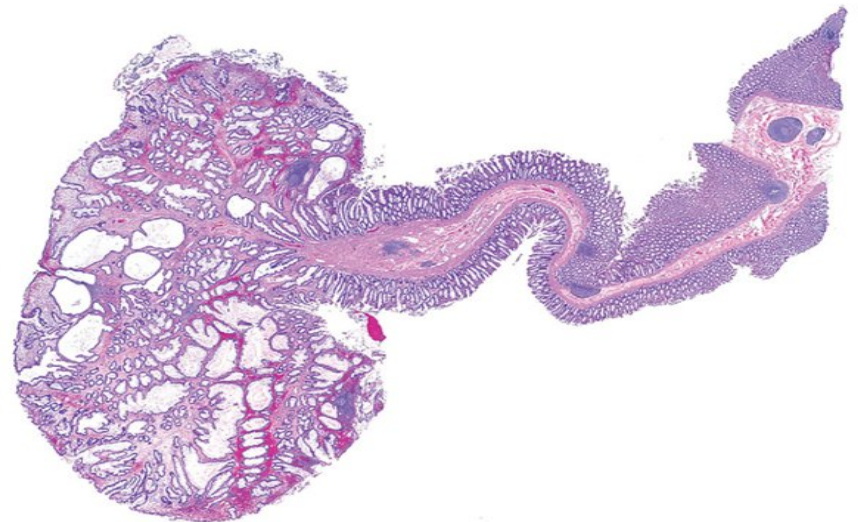
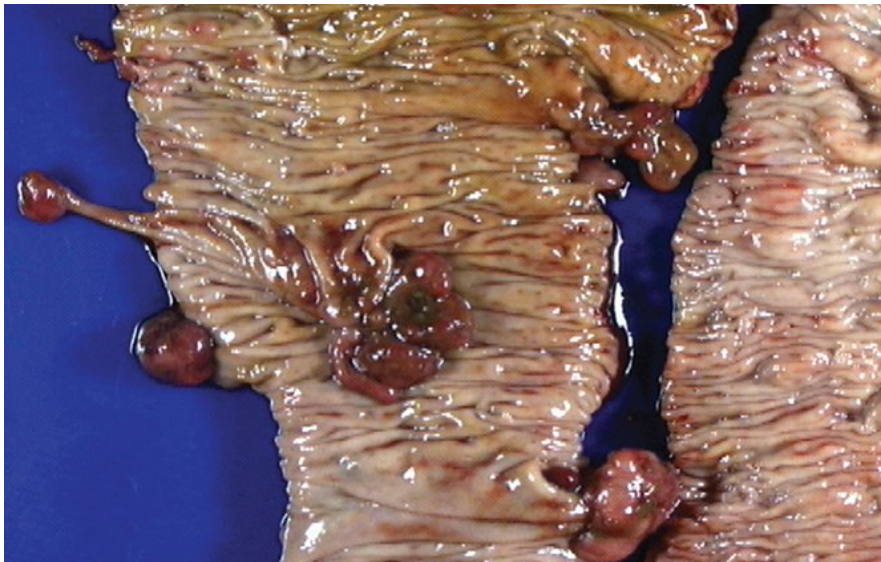


Hamartomatous polyps:

- Hamartomas are disorganized, tumor-like growths composed of mature cell types normally present at the site at which the polyp develops.
- Can be single or multiple.
- Could be sporadic or syndromic.

Juvenile polyps:

- Most common type of hamartomatous polyp.
- Characteristically are located in rectum & present with:
 - Rectal bleeding.
 - Prolapse: the polyp protrudes through anal sphincter.
 - Can be Sporadic or part of Peutz-Jeghers syndrome.
 - Microscopically shows cystically dilated crypts filled with mucin and inflammatory debris, but no dysplasia.



Peutz-Jeghers syndrome:

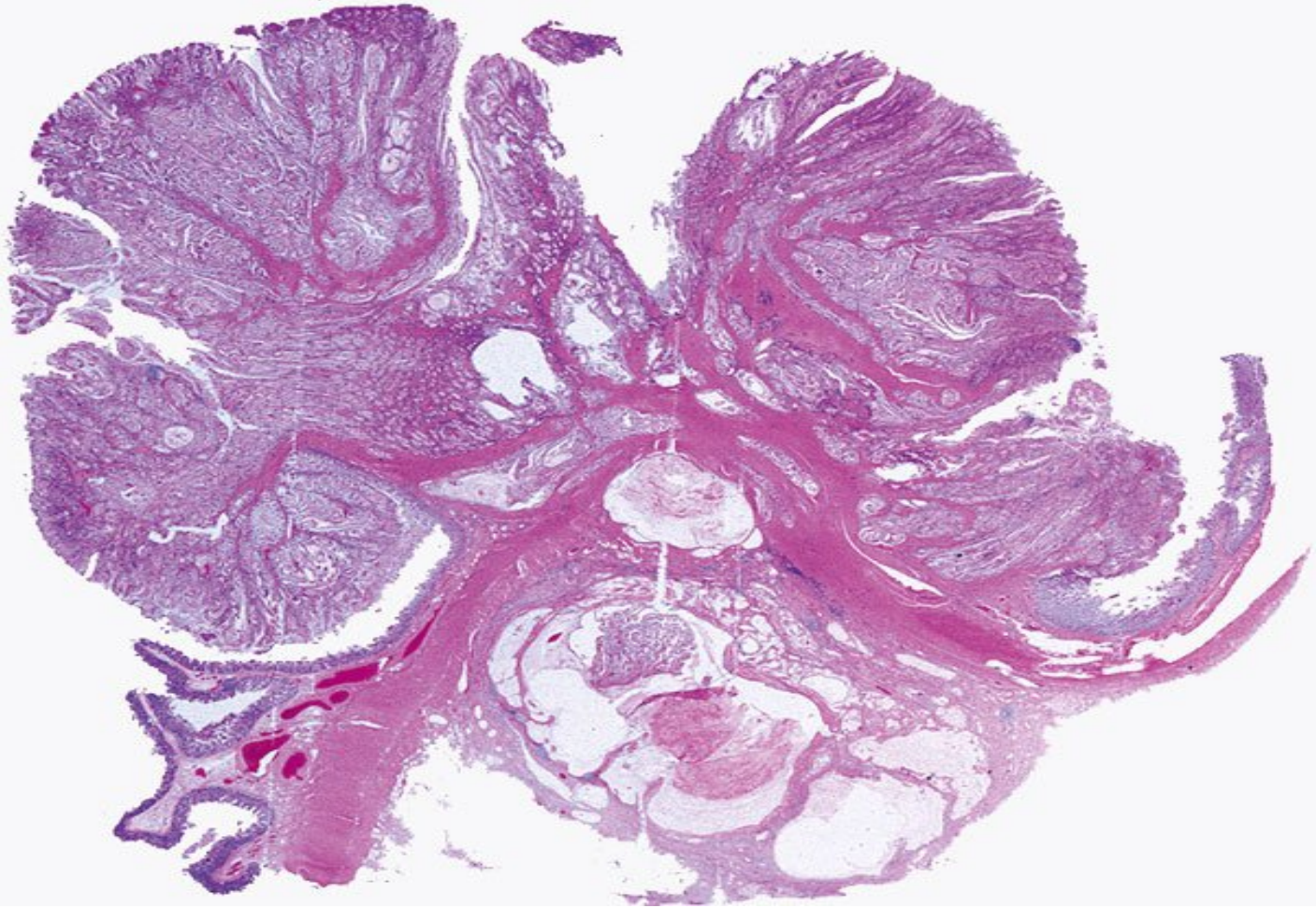
- Rare autosomal dominant disorder.
- Defined by:-
 - Presence of multiple GIT hamartomatous polyps.
 - Mucocutaneous hyperpigmentation.
 - Increased risk of several malignancies e.g. Colon, pancreas, breast, lung, ovaries, uterus, and testes.
- The polyps are most common in small intestine.

PJ syndrome:
Mucocutaneous
hyperpigmentation



PJ syndrome:
hamartomatous
polyps



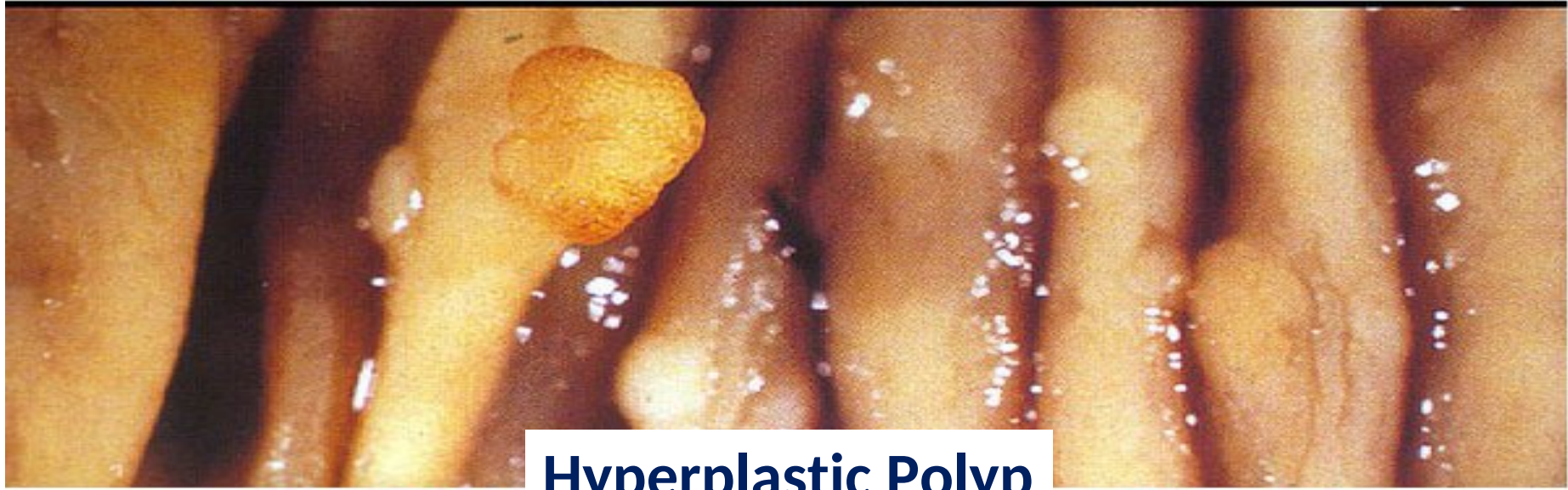


PJ syndrome: hamartomatous polyp

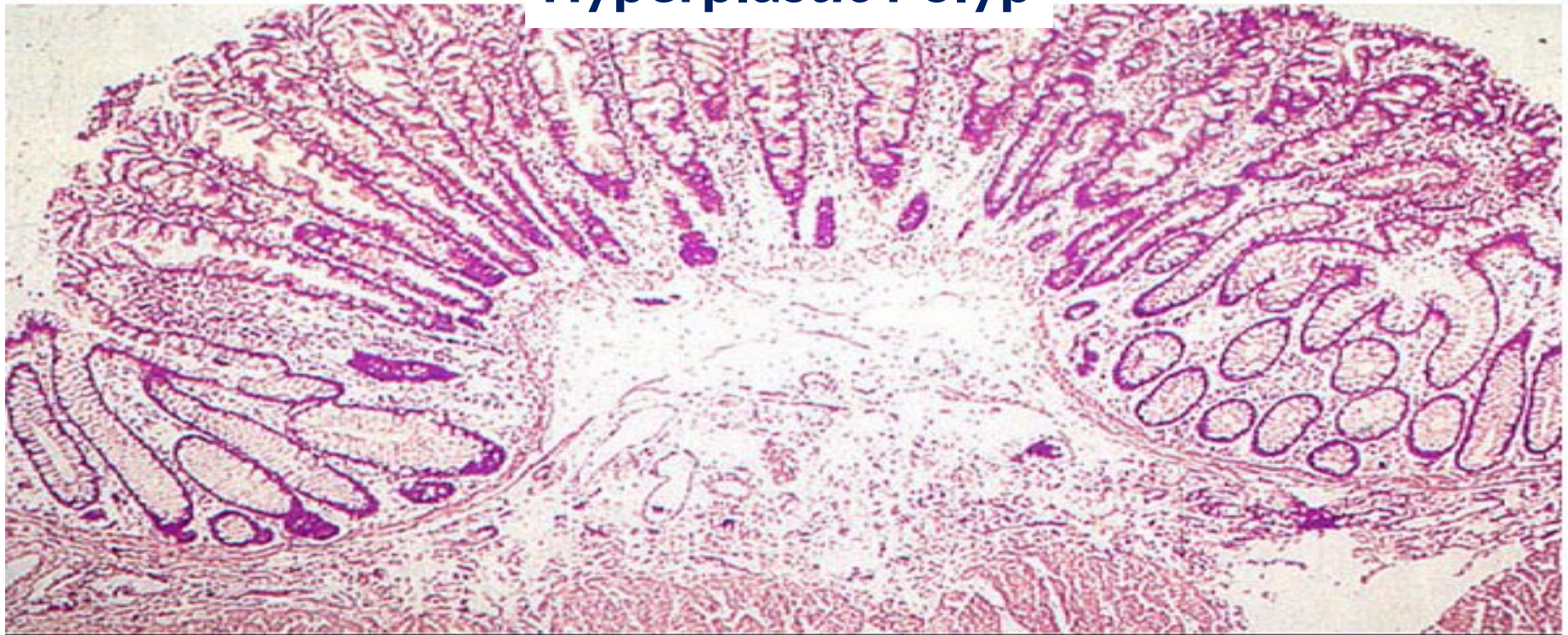
Histology: characteristic arborizing (tree-like) network of (CT, smooth muscle, lamina propria, and glands which are lined by normal-appearing intestinal epithelium).

Hyperplastic Polyps:

- The most commonly encountered polyp in the adult colorectum.
- More common in colon.
- Common in the 6th and 7th decades of life.
- A biopsy of this lesion shows serrated architecture with a sawtooth appearance, with serrations generally limited to the upper half of the crypts, and No evidence of conventional dysplasia.
- The risk to progress to malignancy is questionable.



Hyperplastic Polyp

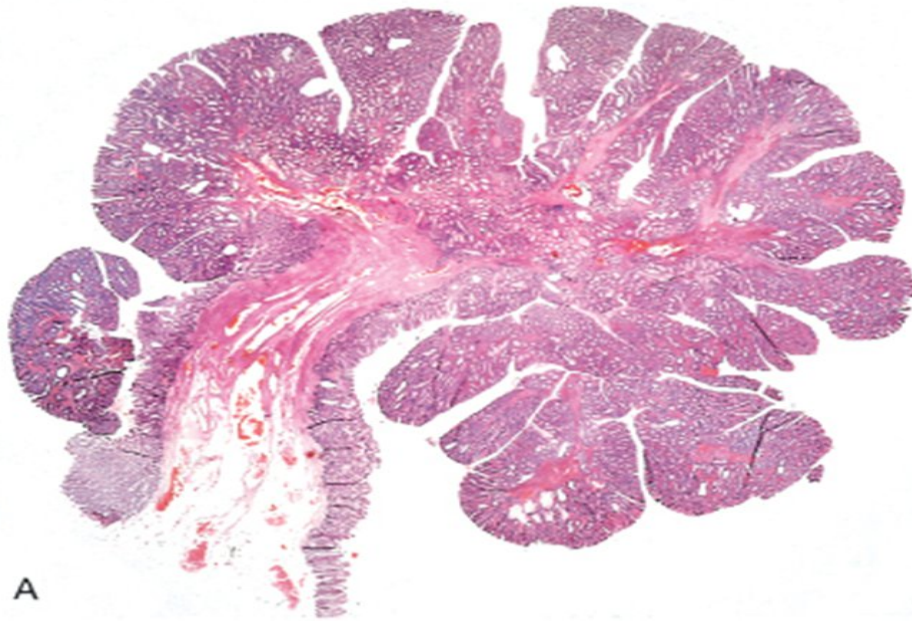


Neoplastic polyps

Adenomatous polyp:

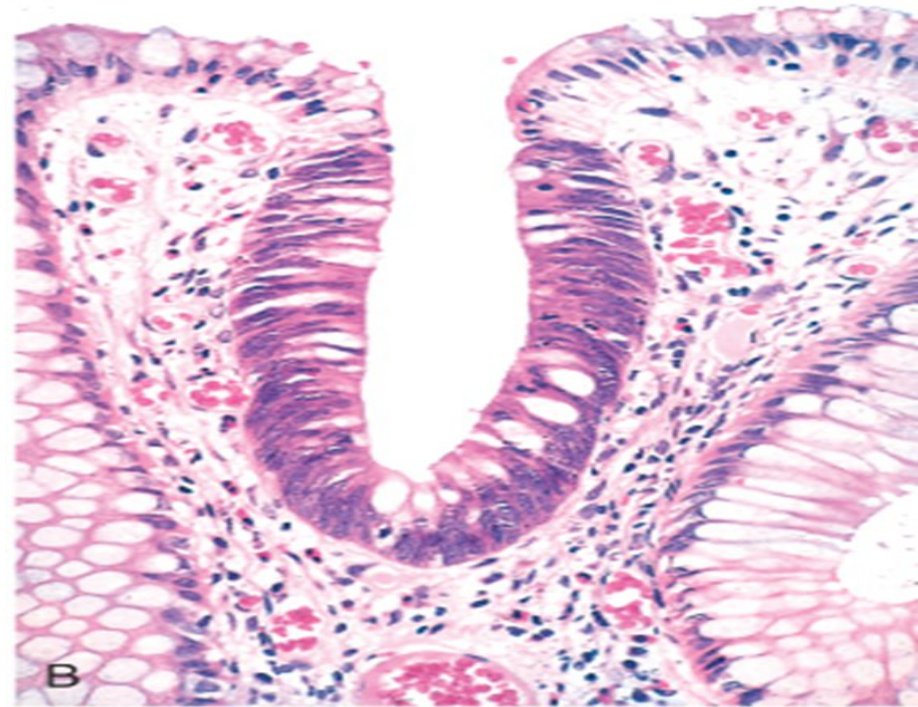
- Also termed adenomas.
- Intestinal adenomas are characterized by the presence of epithelial dysplasia.
- Are benign polyps that give rise to a majority of colorectal adenocarcinomas.
- However, not all adenomas will progress to adenocarcinoma.

- Beginning at age 50.
- Size ranges from 0.3 to 10 cm.
- Can be pedunculated or sessile.
- Histologically: the cytologic hallmark is epithelial dysplasia.



A

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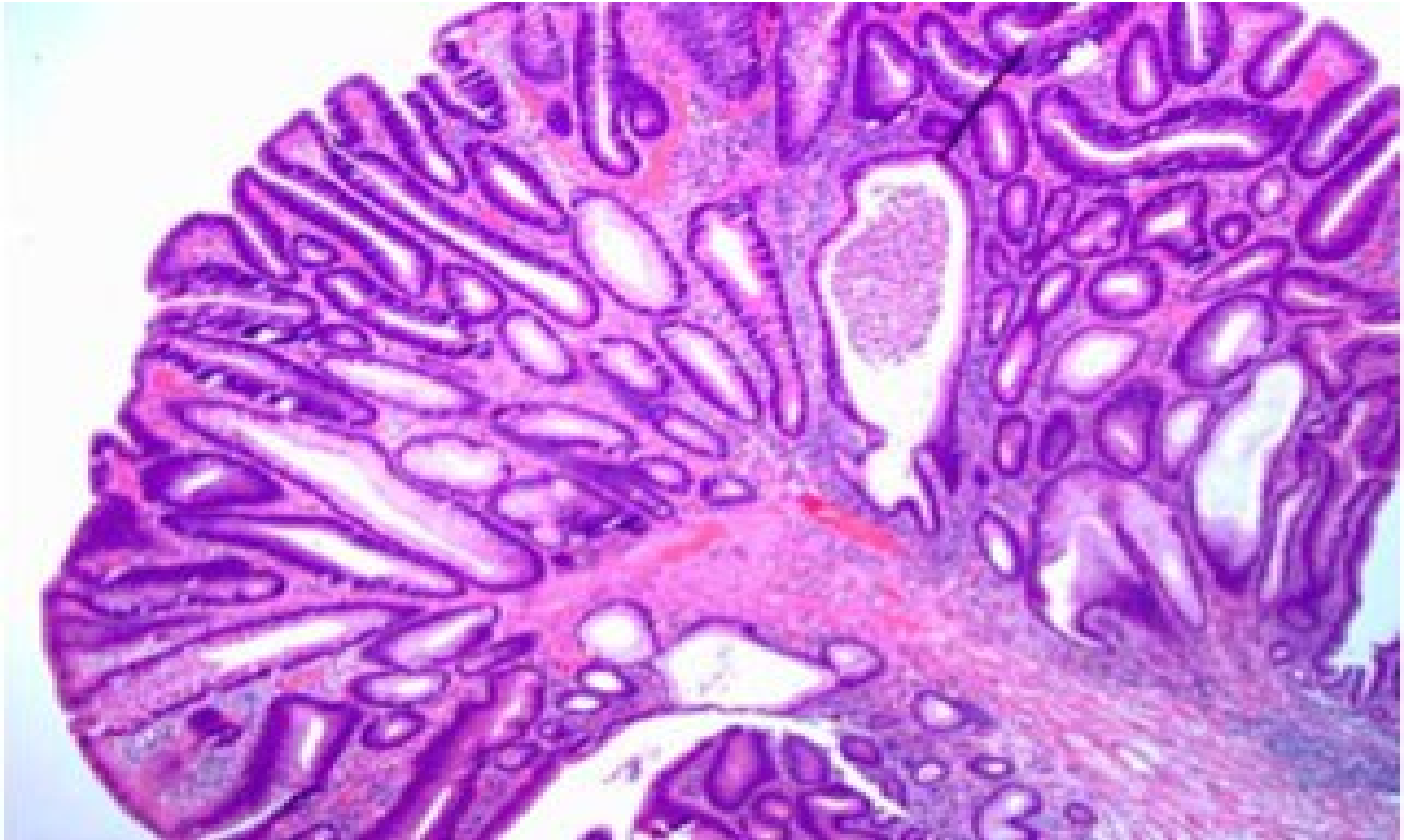
B

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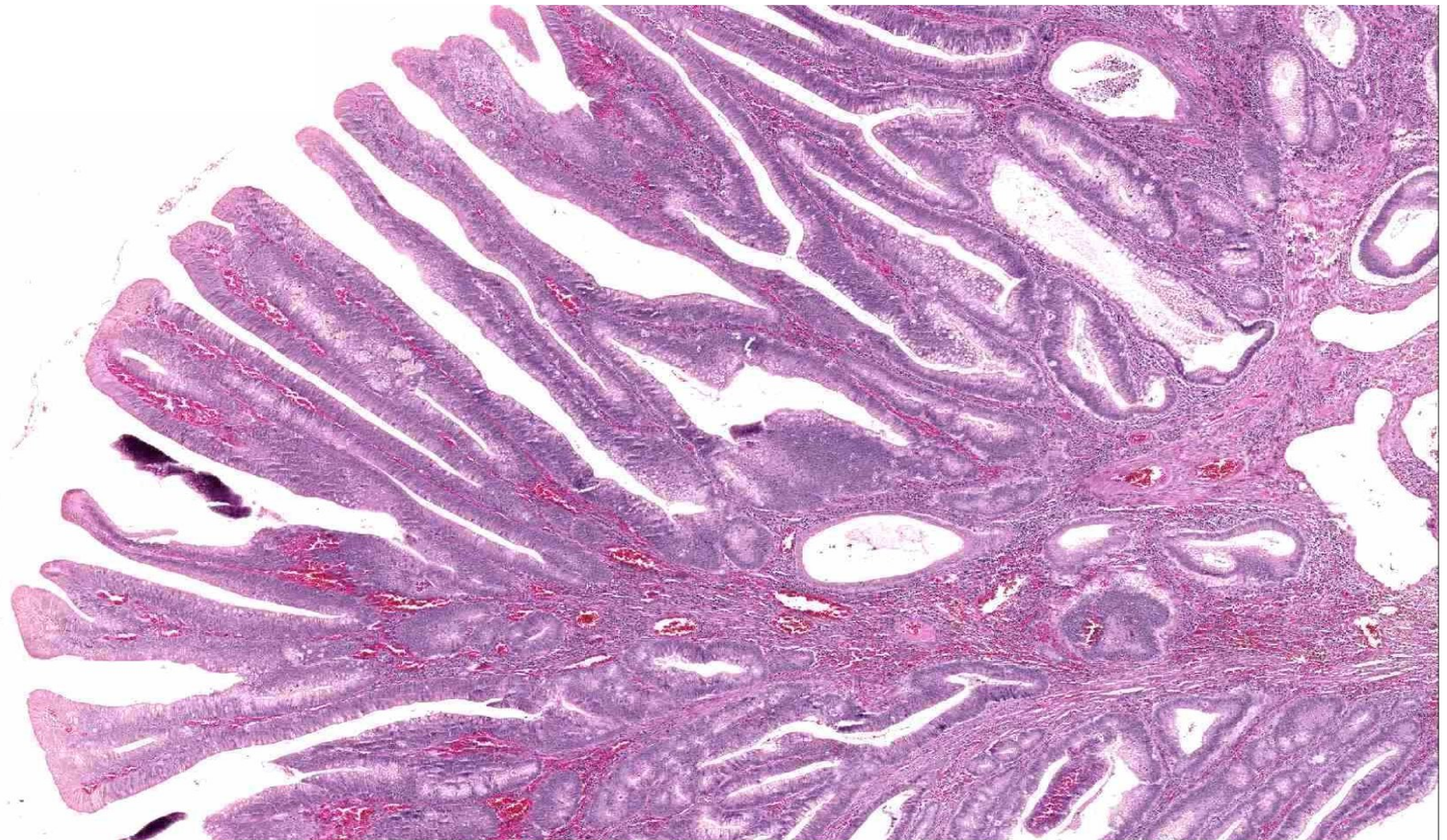
A, Pedunculated adenomatous polyp. **B**, small focus of dysplasia in an otherwise normal (mucin-secreting, clear) colonic mucosa.

- Adenomatous Polyp /Types:
 - **Tubular:** tend to be small pedunculated polyps, composed of small rounded or tubular glands.
 - **Villous:** often larger and sessile, covered by slender villi, more frequently contain foci of invasive malignancy.
 - **Tubulovillous:** adenomas that have a mixture of tubular and villous elements.

Tubular adenoma



Villous Adenoma



Tubulovillous Adenoma

